

Volumes by the Disc and Washer Methods



The disc method (about the x-axis)

- 1 Find the volume of the solid formed by revolving $y = \sqrt{x}$ about the x-axis on $[0, 4]$.
 - 2 Find the volume of the solid formed by revolving $y = x^2$ about the x-axis on $[0, 2]$.
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The disc method (about the y-axis)

- 3 Find the volume formed by revolving $x = y^2$ about the y-axis on $y \in [0, 2]$.
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The washer method

- 4 Find the volume formed by revolving the region between $y = x$ and $y = x^2$ (on $[0, 1]$) about the x-axis.
 - 5 Find the volume formed by revolving the region between $y = \sqrt{x}$ and $y = x$ about the x-axis on $[0, 1]$.
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Volumes about a horizontal or vertical line (not an axis)

- 6 Find the volume formed by revolving the region bounded by $y = x^2$ and $y = 4$ about the line $y = 4$.
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Volumes with known cross-sections

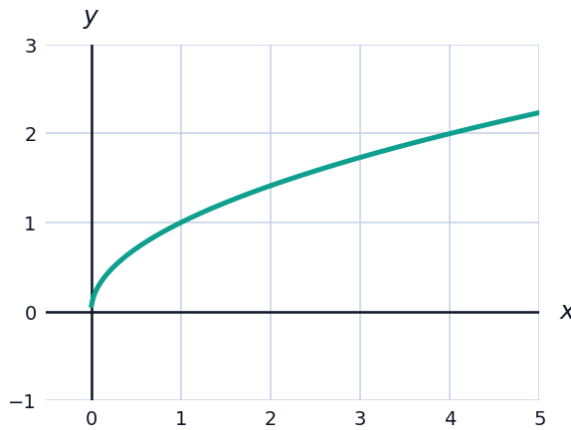
- 7 The base of a solid is the region between $y = x^2$ and $y = 4$ on $[-2, 2]$. Cross-sections perpendicular to the x-axis are squares. Find the volume.
 - 8 The base of a solid is a circle of radius 2 ($x^2 + y^2 = 4$). Cross-sections perpendicular to the x-axis are semicircles. Find the volume.
 - 9 Set up (but do not evaluate) the integral for the volume of the solid whose base is the region bounded by $y = \sin x$ and the x-axis on $[0, \pi]$, with cross-sections perpendicular to the x-axis that are equilateral triangles.
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Volume by revolving about the y-axis (washer method)

- 10 Find the volume formed by revolving the region between $x = y^2$ and $x = 2y$ about the y -axis.
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Visualizing a solid of revolution's generating region

- 11 The graph shows $y = \sqrt{x}$ on $[0, 4]$ from the first disc-method problem. Sketching this region mentally revolved about the x -axis produces a solid — use the volume 8π found earlier to find its average cross-sectional area over the interval, and compare that to the cross-sectional area at the midpoint $x = 2$.



A capstone washer-method problem

- 12 Find the volume formed by revolving the region between $y = 4 - x^2$ and $y = x^2$ about the x -axis (using the region where $4 - x^2 \geq x^2$).
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Volumes with rectangular cross-sections

- 13 The base of a solid is the region bounded by $y = x^2$ and $y = 1$. Cross-sections perpendicular to the x -axis are rectangles with height twice the base. Find the volume.
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A capstone volume application

- 14 A bowl is formed by revolving $y = \frac{x^2}{4}$ (for $0 \leq x \leq 4$, where x is horizontal distance and y is depth from the rim) about the y -axis. Set up (do not evaluate) the disc-method integral for the volume of the bowl in terms of y , using the fact that $x^2 = 4y$.
- 15 Find the volume formed by revolving $y = e^x$ about the x -axis on $[0, 1]$.
- 16 A vase is designed by revolving $y = \frac{1}{2}x + 1$ about the x -axis on $[0, 6]$, where x is the height (inches) and y is the radius. Find the volume of the vase.

17 For the region between $y = x$ and $y = x^3$ on $[0, 1]$ (where $y = x \geq y = x^3$), complete each step for the solid formed by revolving this region about the x -axis:

a Identify the outer radius R and inner radius r .

b Set up the washer-method integral.

c Evaluate the integral.